

1. Equations of a line:
- $\vec{v} = (a, b, c)$
- ,
- $P_0 = (x_0, y_0, z_0)$

$$\vec{r}(t) = P_0 + t\vec{v}$$

$$x = x_0 + ta, \quad y = y_0 + tb, \quad z = z_0 + tc$$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

2. Equation of a plane:
- $P_0 = (x_0, y_0, z_0)$
- ,
- $\vec{n} = (n_x, n_y, n_z)$

$$n_x(x - x_0) + n_y(y - y_0) + n_z(z - z_0) = 0$$

3. Displacement vector:
- $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

4. Velocity vector:
- $\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$

5. Speed:
- $|\vec{v}| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$

6. Arc length parameter:
- $s(t) = \int_{t_0}^t |\vec{v}(\tau)| d\tau$

7. Length of a curve
- C
- :
- $\vec{r}(t)$
- ,
- $\alpha \leq t \leq \beta$
- ,

$$L = \int_{t=\alpha}^{t=\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

8. Unit tangent vector:
- $\hat{T} = \frac{d\vec{r}}{ds} = \frac{\vec{v}}{|\vec{v}|}$

9. Curvature:
- $\kappa(t) = \left| \frac{d\hat{T}}{ds} \right| = \frac{1}{|\vec{v}|} \left| \frac{d\hat{T}}{dt} \right| = \frac{|\dot{\vec{r}} \times \ddot{\vec{r}}|}{|\dot{\vec{r}}|^3}$

10. Principal unit normal:
- $\hat{N} = \frac{1}{\kappa} \frac{d\hat{T}}{ds} = \frac{\frac{d\hat{T}}{dt}}{\left| \frac{d\hat{T}}{dt} \right|}$

11. Binormal vector:
- $\hat{B} = \hat{T} \times \hat{N}$

12. Acceleration vector:
- $\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}$

13. Tangential and normal components of the acceleration vector:

$$\vec{a} = a_T \hat{T} + a_N \hat{N},$$

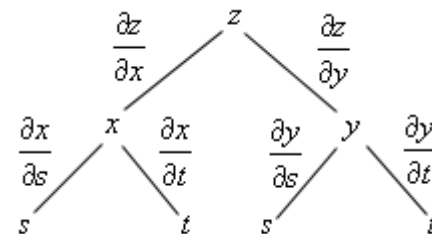
$$a_T = \frac{d}{dt} |\vec{v}|, \quad a_N = \kappa |\vec{v}|^2 = \sqrt{|\vec{a}|^2 - a_T^2}$$

14. Chain rule for multivariable functions
- ¹
- : Given
- $z = f(x, y)$
- ,
- $x = g(s, t)$
- ,
- $y = h(s, t)$
- ,

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

The tree diagram for this case is,



15. Gradient of a function
- $f(x, y, z)$
- :

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

16. Directional derivative along the unit vector
- $\hat{u}$
- at point
- P_0
- :

$$\left(\frac{df}{ds} \right)_{\hat{u}, P_0} = D_{\hat{u}} f|_{P_0} = \vec{\nabla} f \cdot \hat{u}|_{P_0}$$

¹ These chain rule formulas and tree can be easily generalized to a function of “n” variables, with each one of these variables being in turn functions of “m” variables.